

Omkar Kulkarni

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SUMMARY

PhD researcher who designs and implements AI algorithms and methods for Medical Image Analysis. Demonstrated proficiency working with weakly supervised segmentation and 3D modeling from medical image data. Currently exploring medical image denoising to mitigate distribution shift in cross-site medical image data. Published at AIME and IEEE AIMHC.

PUBLICATIONS

- “Learning to Segment using Summary Statistics and Weak Supervision” In *The 24th International Conference on Artificial Intelligence in Medicine, 2026*. DOI:10.1007/978-3-032-30710-1_2 Arxiv: [2605.03059](https://arxiv.org/abs/2605.03059)
- “3D Reconstruction of the Carotid Artery from Handheld Ultrasound Videos” In *Proceedings of the First IEEE Conference on Artificial Intelligence for Medicine, Health, and Care, 2024*. DOI:10.1109/AIMHC59811.2024.00048

EDUCATION

University of Maryland Baltimore County (UMBC)

Doctor of Philosophy (Ph.D) in Computer Science

Baltimore, MD

Dec 2028 (Expected)

Birla Institute of Technology and Science, Pilani (BITS)

Bachelor of Engineering (B.E) in Computer Science

Goa, India

May 2018

EXPERIENCE

Graduate Assistant

University of Maryland Baltimore County, CoRaL lab

Baltimore, MD

Jan 2022 – Present

- Image Denoising: Investigating self-supervised denoising techniques for medical images to mitigate distribution shift across multi-site imaging data, with the goal of improving downstream segmentation robustness in cross-site settings
- Weakly Supervised Learning: Designed a weak supervision learning framework to reduce the reliance on hand-annotated data for training segmentation models for ultrasound and CT scan data. Our research yields near full-supervision segmentation performance on two publicly available datasets while significantly reducing annotation time and effort. Published at AIME 2026
- Medical Image Segmentation: Achieved a 94% dice segmentation accuracy for carotid arteries in handheld ultrasound videos through the development of a custom deep learning model
- 3D Reconstruction: Improved clinical visualization of the carotid artery by transforming 2D segmentation masks from ultrasound into 3D models. Published at AIMHC 2024
- Key Tools: Python, PyTorch, Optuna, VGG/VIA annotation, Git

Software Developer II

Innova Solutions/Transplace 3PL

Hyderabad, INDIA

July 2018 – July 2020

- Architecture Implementation: Reduced average per-client onboarding time by ~24 hours by co-developing a templated validation framework, translating architectural designs into production code
- Feature Engineering: Enhanced system stability in a high-traffic 3PL environment by developing core shipment features and resolving critical software defects
- Key Tools: Java, OOSD, Angular, SQL, Git

Research Exchange Student

Singapore University of Technology and Design

Singapore

Jan 2018 – June 2018

- Predictive Modeling: Engineered a hit-prediction model achieving 67.17% accuracy by extracting and analyzing acoustic descriptors from a custom-built dataset via the Spotify Web API
- Key Tools: Python, Scikit-learn, Keras

Computer Vision Intern

Konnet Solutions

Pune, INDIA

May 2017 – July 2017

- Dataset Generation: Curated a dataset of vehicle license plates captured at highway and main road toll booths consisting of over 3,000 images.
- Optical Character Recognition: Automated vehicle license plate identification by achieving 87% character-level accuracy by developing a deep learning-based OCR license plate reader using the curated dataset.
- Key Tools: Python, TensorFlow, OpenCV

TECHNICAL PROFICIENCY

Languages: Python, C/C++, Java, SQL

AI/ML Frameworks: PyTorch, TensorFlow, Keras, Optuna, Scikit-learn, OpenCV

Research Specializations: Self-Supervised Learning (SSL), Weakly Supervised Learning, Medical Image Analysis, 3D Reconstruction

Infrastructure & Tools: SLURM, Linux/Unix, Git, Cloud/Cluster Computing, Distributed Computing, VGG/VIA (Annotation), Angular, OOSD, All Of Us researcher workbench

PROFESSIONAL/COMMUNITY SERVICE

Reviewer

- Journal of Imaging Informatics in Medicine (JIIM), SIIM, July 2026
- Reproducible AI Workshop, AAAI 2026

Graduate Teaching Assistant, UMBC

Fall 2021, Spring 2025 – Present

- Courses: Machine Learning, Natural Language Processing, Practical Deep Learning, Computer Organization/Architecture

Graduate Student Association, UMBC

2021-2023

- **Leadership experience:** Senator, Chair of the Equity of Funds committee, Grants Review Panel member

Technology Incubator Program, BITS

2017

- Mentored a batch of ~25 undergraduate students to build their own Optical Character Recognition model using the SVHN dataset.

AWARDS, ACHIEVEMENTS, HONORS

Doctoral Consortium Presenter, AIME 2026

July 2026

Selected to present doctoral research proposal at the 24th International Conference on AI in Medicine.

Professional Development Grant, UMBC GSA

Feb 2024, July 2026

A travel grant to support expenses for presenting my research.

Member, Sigma Xi, The Scientific Research Honor Society

Mar 2026

Nominated by the National Committee based on demonstrated research aptitude in AI-driven medical image analysis

INSPIRE Scholarship for higher education

June 2012

A state level scholarship awarded to students graduating high school within the top 1 percentile.